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Associations between Aging and Vitamin D Status with Whole-Body Nitric Oxide Production and Markers of Endothelial Function

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Abstract

Background Aging and vitamin D deficiency have been associated with reduced nitric oxide (NO) synthesis and impaired endothelial function (EF) but the evidence in humans remains weak. **Objectives** Two independent cross-sectional studies were designed to evaluate the association between age, sex, and plasma vitamin D concentrations with physiological and biochemical biomarkers of NO synthesis and EF in young and older healthy participants (Study 1) and in overweight and obese postmenopausal females (Study 2). **Methods** In Study 1, 40 young (20–49 y) and older (50–75 y) males and females (10 participants per age and sex group) were included. Resting blood pressure and ear-to-finger peripheral pulse wave velocity (PWV) were measured. A stable-isotopic method was used to determine whole-body NO production. Plasma 25-hydroxyvitamin D (25(OH)D), nitrate, nitrite, and asymmetric dimethylarginine (ADMA) concentrations were determined. In Study 2, 80 older overweight and obese females (age 61.2 ± 6.2 y, body mass index 29.5 ± 4.4 kg/m²) were recruited. Postocclusion reactive hyperemia (PORH) and peripheral PWV were measured. Plasma concentrations of 25(OH)D, nitrate, cyclic guanosine monophosphate, 3-nitrotyrosine (3-NT), endothelin-1, vascular endothelial growth factor, and ADMA were determined. **Results** In Study 1, whole-body NO production was significantly greater in young compared with older participants (0.61 ± 0.30 $\mu\text{mol}\cdot\text{h}^{-1}\cdot\text{kg}^{-1}$ compared with 0.39 ± 0.10 $\mu\text{mol}\cdot\text{h}^{-1}\cdot\text{kg}^{-1}$, $P = 0.01$) but there was no evidence of a sex difference ($P = 0.81$). Plasma 25(OH)D concentration was not associated with PWV ($r = 0.18$, $P = 0.28$) or whole-body NO production ($r = -0.20$, $P = 0.22$). Plasma ADMA concentration was associated positively with age ($r = 0.35$, $P = 0.03$) and negatively with whole-body NO production ($r = -0.33$, $P = 0.04$). In Study 2, age was associated with lower PORH ($r = -0.28$, $P = 0.02$) and greater ADMA concentrations ($r = 0.22$, $P = 0.04$). Plasma 25(OH)D concentration was inversely associated with 3-NT concentrations ($r = -0.31$, $P = 0.004$). **Conclusions** Older age was associated with lower whole-body NO production. Plasma vitamin D concentrations were not associated with NO production or markers of EF but showed a weak, significant correlation with oxidative stress in postmenopausal overweight females.

Original language English

Pages (from-to) 469-478

Number of pages 10

Journal [Journal of Nutrition](#)

Volume 154

Issue number 2

Early online date 3 Dec 2023

DOIs • <https://doi.org/10.1016/j.tjnut.2023.12.002>

Publication status Published - Feb 2024

Bibliographical note

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Keywords

- ageing
- endothelial function
- nitric oxide
- pulse wave velocity
- vitamin D

Access to Document

- [10.1016/j.tjnut.2023.12.002](https://doi.org/10.1016/j.tjnut.2023.12.002)

Other files and links

- [Link to publication in Scopus](#)

Embargoed Document

- Vitamin D status in overweight and obese Endothelial Function_2023

Accepted author manuscript, 596 KB

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Siervo, M., Hussin, A. M., Calella, P., Ashor, A., Shannon, O. M., Mendes, I., Stephan, B. C., [Zheng, D.](#), Hill, T., & Mathers, J. C. (2024). [Associations between Aging and Vitamin D Status with Whole-Body Nitric Oxide Production and Markers of Endothelial Function.](#) **Journal of Nutrition**, **154**(2), 469-478. <https://doi.org/10.1016/j.tjnut.2023.12.002>
[Associations between Aging and Vitamin D Status with Whole-Body Nitric Oxide Production and Markers of Endothelial Function.](#) / Siervo, Mario; Hussin, Azizah Mat; Calella, Patrizia et al. In: [Journal of Nutrition](#), Vol. 154, No. 2, 02.2024, p. 469-478.

Research output: Contribution to journal > Article > peer-review

Siervo, M, Hussin, AM, Calella, P, Ashor, A, Shannon, OM, Mendes, I, Stephan, BC, [Zheng, D](#), Hill, T & Mathers, JC 2024, '[Associations between Aging and Vitamin D Status with Whole-Body Nitric Oxide Production and Markers of Endothelial Function](#)', **Journal of Nutrition**, vol. 154, no. 2, pp. 469-478. <https://doi.org/10.1016/j.tjnut.2023.12.002>

Siervo M, Hussin AM, Calella P, Ashor A, Shannon OM, Mendes I et al. [Associations between Aging and Vitamin D Status with Whole-Body Nitric Oxide Production and Markers of Endothelial Function.](#) *Journal of Nutrition*. 2024 Feb;154(2):469-478. Epub 2023 Dec 3. doi: 10.1016/j.tjnut.2023.12.002

Siervo, Mario ; Hussin, Azizah Mat ; Calella, Patrizia et al. / [Associations between Aging and Vitamin D Status with Whole-Body Nitric Oxide Production and Markers of Endothelial Function.](#) In: *Journal of Nutrition*. 2024 ; Vol. 154, No. 2. pp. 469-478.

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keywords = "ageing, endothelial function, nitric oxide, pulse wave velocity, vitamin D",

author = "Mario Siervo and Hussin, {Azizah Mat} and Patrizia Calella and Ammar Ashor and Shannon, {Oliver M} and Ines Mendes and Stephan, {Blossom Cm} and Dingchang Zheng and Tom Hill and Mathers, {John C}",
note = "Copyright {\textcopyright} 2023. Published by Elsevier Inc.",

year = "2024",

month = feb,

doi = "10.1016/j.tjnut.2023.12.002",

language = "English",

volume = "154",

pages = "469--478",

journal = "Journal of Nutrition",

issn = "0022-3166",

publisher = "American Society for Nutrition",

number = "2",

}

TY - JOUR

T1 - Associations between Aging and Vitamin D Status with Whole-Body Nitric Oxide Production and Markers of Endothelial Function

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PY - 2024/2

Y1 - 2024/2

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KW - pulse wave velocity

KW - vitamin D

UR - <http://www.scopus.com/inward/record.url?scp=85180561303&partnerID=8YFLogxK>

U2 - 10.1016/j.tjnut.2023.12.002

DO - 10.1016/j.tjnut.2023.12.002

M3 - Article

C2 - 38048992

SN - 0022-3166

VL - 154

SP - 469

EP - 478

JO - Journal of Nutrition

JF - Journal of Nutrition

IS - 2

ER -



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